

Computational Physics Classes No.8 - Report

Molecular dynamics simulations in 2D

Natalia Kowalczyk

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Abstract

Molecular dynamics simulations of a two-dimensional system of interacting particles were performed using the Lennard–Jones potential and the velocity–Verlet integration scheme. Periodic boundary conditions and a Nose–Hoover thermostat were applied to study microcanonical and canonical ensembles. Energy conservation, thermalization, velocity distributions, and crystallization during cooling were investigated. The numerical results reproduce the expected physical behavior and agree with analytical predictions.

1 Introduction

Molecular dynamics (MD) is a numerical technique used to study the time evolution of interacting many-particle systems by solving classical equations of motion. In this work, a two-dimensional system of argon atoms confined in a square box is investigated. Interatomic interactions are described by the Lennard–Jones potential, while periodic boundary conditions are applied to mimic an infinite system.

The goal of this project is to implement a stable and physically correct MD simulation and to verify its correctness by examining energy conservation, thermalization efficiency, velocity distributions, and structural ordering during cooling.

2 Model and Numerical Methods

2.1 Physical Model

The simulated system consists of N identical particles representing argon atoms confined in a two-dimensional square box of size $L \times L$. The dynamics of the system are governed by classical equations of motion, with interparticle interactions described by the Lennard–Jones (LJ) potential. This potential captures the essential physical features of neutral atomic interactions: a strong short-range repulsion originating from the Pauli exclusion principle and a weaker long-range attraction due to van der Waals forces.

The pairwise interaction potential between particles i and j separated by a distance r_{ij} is given by

$$U_{\text{LJ}}(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right], \quad (1)$$

where ϵ sets the characteristic energy scale of the interaction and σ corresponds to the effective particle diameter.

To limit the computational cost, a cutoff radius r_{cut} is introduced, such that interactions are neglected for particle separations $r_{ij} > r_{\text{cut}}$. In order to avoid artificial discontinuities of the force at the cutoff distance, a shifted–force version of the Lennard–Jones potential is employed. This modification ensures that both the potential energy and the force smoothly vanish at $r = r_{\text{cut}}$, which is crucial for numerical stability and accurate energy conservation.

All simulations are performed using reduced (effective) units natural for argon atoms. The particle mass, Lennard–Jones parameters, and Boltzmann constant are set to unity,

$$m = 1, \quad \epsilon = 1, \quad \sigma = 1, \quad k_B = 1, \quad (2)$$

which simplifies the equations of motion and energy expressions. Temperatures expressed in Kelvin are converted to reduced units using the reference temperature $T_{\text{ref}} = 119 \text{ K}$. When necessary, physical units can be recovered by multiplying reduced quantities by the corresponding reference values.

2.2 Boundary Conditions

To mimic an infinite system and reduce finite-size effects associated with the limited number of particles, periodic boundary conditions are applied in both spatial directions. Under these conditions, when a particle crosses one boundary of the simulation box, it re-enters the box from the opposite side while preserving its velocity.

Interparticle distances are calculated using the minimum image convention, which guarantees that each particle interacts only with the nearest periodic image of every other particle. For two particles separated by coordinate differences Δx and Δy , the effective displacement components are defined as

$$\Delta x \rightarrow \Delta x - L \cdot \text{round}\left(\frac{\Delta x}{L}\right), \quad \Delta y \rightarrow \Delta y - L \cdot \text{round}\left(\frac{\Delta y}{L}\right), \quad (3)$$

so that the resulting distance r_{ij} corresponds to the shortest separation across periodic boundaries. This procedure is essential for the correct implementation of the Lennard-Jones interaction in a periodically repeated system.

2.3 Time Integration and Thermostat

The time evolution of the system is obtained by numerically integrating the classical equations of motion using the velocity-Verlet algorithm. This integration scheme is symplectic and time-reversible, which ensures good long-term stability and makes it particularly suitable for molecular dynamics simulations. In each time step, particle velocities are first advanced by a half step, followed by a full update of particle positions. After recalculating forces and accelerations at the new positions, the velocities are completed to the full time step.

To control the temperature and allow simulations in the canonical ensemble, a Nose-Hoover thermostat is employed. This method introduces an additional dynamical variable that represents coupling to an external heat bath. The thermostat enables exchange of energy between the system and the reservoir, causing the instantaneous temperature to fluctuate around a prescribed target value.

The strength of the coupling between the system and the heat bath is controlled by the parameter Q . Larger values of Q correspond to weak coupling and large temperature fluctuations, while smaller values enforce stronger coupling and suppress fluctuations. The Nose-Hoover equations are solved self-consistently at each time step using a small number of Picard iterations, which ensures stable thermalization without significantly distorting the particle dynamics.

3 Simulation Parameters

All simulations were performed using the following parameters, as specified in the project instructions:

- Number of particles: $N = 201$
- Simulation box size: $L = 25$
- Time step: $\Delta t = 0.002$
- Lennard-Jones cutoff radius: $r_{\text{cut}} = 2.7$
- Reference temperature: $T_{\text{ref}} = 119 \text{ K}$

Unless stated otherwise, temperatures quoted in the simulations are expressed in reduced units obtained by dividing the physical temperature by T_{ref} . These parameters were chosen to ensure numerical stability while allowing direct comparison with the example results provided in the project description.

4 Results

4.1 Initial Conditions

The initial particle positions were generated by placing the atoms at the nodes of a regular square lattice within the simulation box. This procedure prevents unphysically small interparticle distances that could lead to extremely large forces and numerical instability at the beginning of the simulation. The resulting spatial distribution is uniform and covers the entire box, as shown in Fig. 1.

Initial velocities were assigned according to the Maxwell–Boltzmann distribution corresponding to the target temperature. The velocity components were drawn independently from normal distributions with zero mean. Due to the finite number of particles, the initial center-of-mass velocity was nonzero and was therefore removed explicitly. As a consequence, the instantaneous temperature calculated immediately after drift removal differed slightly from the target value. A final rescaling of velocities was performed to restore the desired temperature, after which the instantaneous temperature matched the prescribed value within numerical precision.

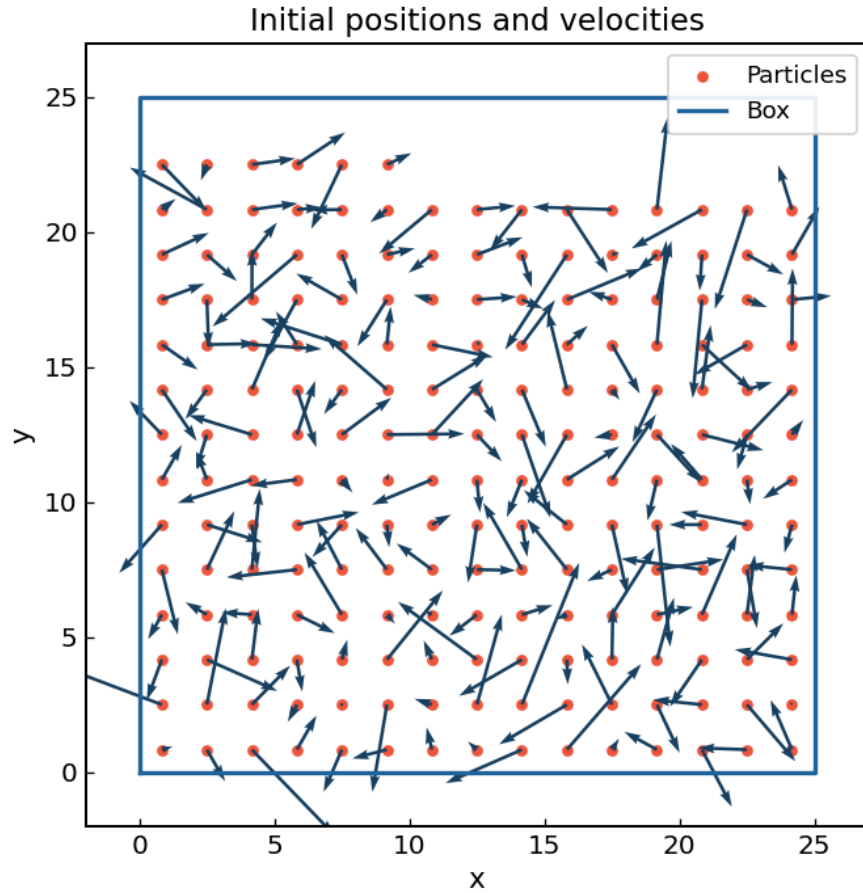


Figure 1: Initial particle positions arranged on a regular lattice. Arrows indicate the initial velocity vectors drawn from the Maxwell–Boltzmann distribution.

4.2 Energy Conservation

To verify the correctness of the velocity–Verlet integration scheme, simulations were performed in the microcanonical (NVE) ensemble with the thermostat effectively turned off. The kinetic energy, potential energy, and total energy were monitored throughout the simulation.

As shown in Fig. 2, the total energy remains constant over time, confirming proper energy conservation. In contrast, the kinetic and potential energies exhibit pronounced fluctuations and are strongly

anticorrelated, as expected for a closed Hamiltonian system. Small residual fluctuations of the total energy originate from the finite time step used in the numerical integration.

In addition, the trajectory of a single particle initially located near the center of the simulation box was recorded. The resulting path is highly irregular and exhibits diffusive behavior caused by frequent collisions with other particles. Due to periodic boundary conditions, the particle repeatedly traverses the simulation box.

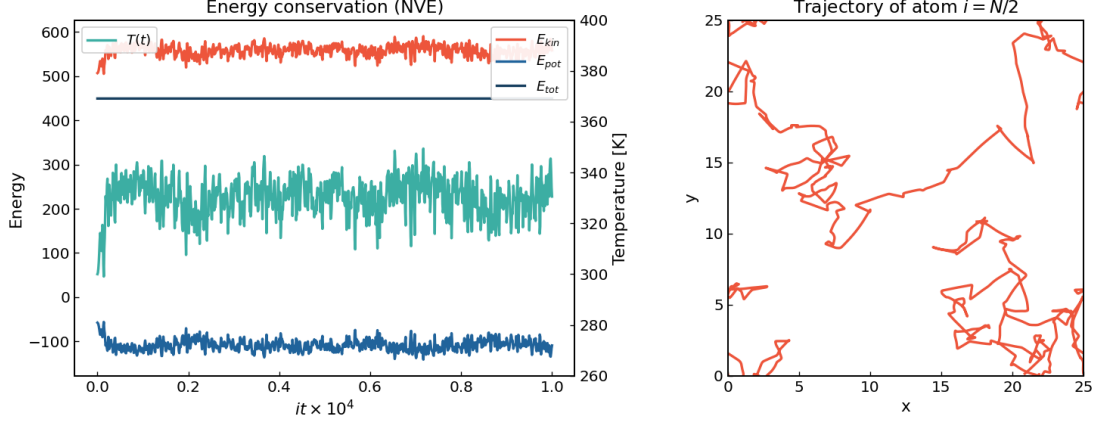


Figure 2: Left: kinetic, potential, and total energy as functions of iteration in the NVE ensemble. Right: trajectory of a single particle showing chaotic, diffusive motion.

4.3 Thermalization and Thermostat Coupling

The efficiency of thermalization was examined using the Nose-Hoover thermostat for different values of the coupling parameter Q . Simulations were performed for $Q = 1, 0.1$, and 0.01 , corresponding to progressively stronger coupling between the system and the heat bath.

Figure 3 shows the time evolution of energies and instantaneous temperature for each case. For $Q = 1$, the temperature exhibits large fluctuations, indicating weak coupling. As Q is reduced, temperature fluctuations decrease and the kinetic energy becomes increasingly stable. For the strongest coupling ($Q = 0.01$), both temperature and kinetic energy remain nearly constant, closely tracking the prescribed target temperature.

The mean temperature obtained by averaging instantaneous values over time agrees with the target temperature for all values of Q , confirming correct implementation of the thermostat.

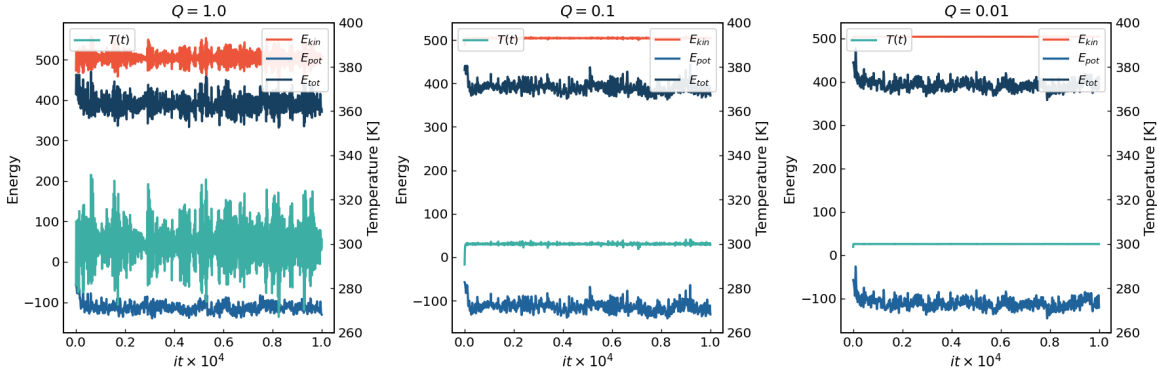


Figure 3: Energies and instantaneous temperature as functions of iteration for different Nose-Hoover coupling parameters Q . Smaller values of Q result in reduced temperature fluctuations.

4.4 Velocity Distribution

The velocity distribution of particles was analyzed for a system coupled weakly to the thermostat ($Q = 10$). Velocity histograms were constructed for three different simulation lengths: $N_{it} = 10^3$, 10^4 , and 10^5 . The histograms were normalized and compared with the analytical Maxwellian velocity distribution for a two-dimensional gas.

As shown in Fig. 4, increasing the number of iterations significantly improves the agreement between numerical and analytical results. For short simulations, noticeable statistical fluctuations are present, while longer simulations yield smooth distributions that closely follow the theoretical curve. Small residual discrepancies persist even for long simulations due to interparticle interactions, as the system is not an ideal gas.

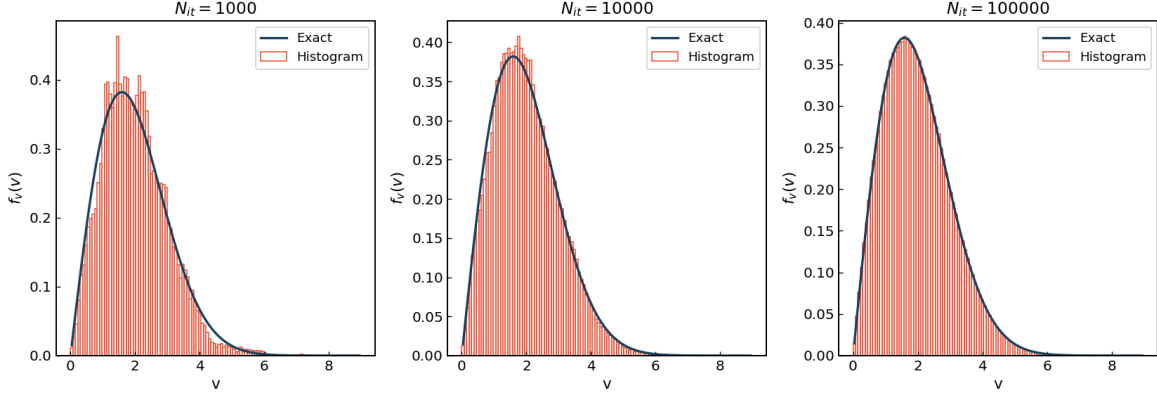


Figure 4: Velocity histograms compared with the analytical Maxwellian distribution for increasing numbers of simulation steps. Statistical convergence improves with longer simulations.

4.5 Cooling and Crystallization

The crystallization process was studied by gradually decreasing the target temperature from $T_{\max}$ to $T_{\min}$ using a stepwise cooling schedule. Two simulations were performed: a fast cooling run with $N_{it} = 10^4$ and a slow cooling run with $N_{it} = 10^5$.

The evolution of instantaneous temperature for both cases is shown in Fig. 5. In the fast cooling regime, the system does not have sufficient time to relax and becomes trapped in metastable configurations, resulting in the formation of multiple small, disordered clusters. In contrast, slow cooling allows the particles to explore phase space more efficiently, leading to the emergence of a more regular structure resembling a triangular lattice.

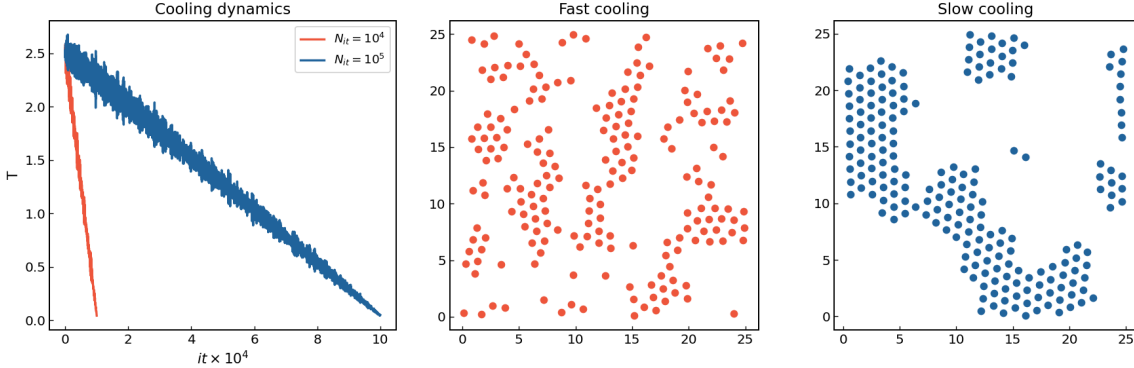


Figure 5: Left: temperature evolution during fast and slow cooling. Middle: final particle positions after fast cooling. Right: final configuration after slow cooling, showing increased structural order.

5 Conclusions

A complete molecular dynamics simulation of a two-dimensional system of interacting particles was successfully implemented and analyzed. The use of the Lennard–Jones potential combined with periodic boundary conditions and the velocity–Verlet integration scheme provided numerically stable and physically consistent particle dynamics.

Energy conservation in the microcanonical ensemble was verified, confirming the correctness of the integration algorithm. The Nose–Hoover thermostat was shown to effectively control the system temperature, with the strength of coupling parameter Q governing the magnitude of temperature and energy fluctuations. Velocity distributions obtained from the simulations converge toward the analytical Maxwellian form with increasing simulation time, demonstrating proper thermal equilibrium behavior.

Cooling simulations revealed a strong dependence of the final structural arrangement on the cooling rate. Fast cooling led to disordered, metastable configurations, while slow cooling allowed the system to relax toward more ordered, crystalline structures. These results are consistent with physical expectations and illustrate the capability of molecular dynamics simulations to capture both equilibrium and non-equilibrium phenomena.

Overall, the obtained results confirm the correctness of the numerical implementation and demonstrate the effectiveness of molecular dynamics methods in studying thermodynamic and structural properties of interacting particle systems.

References

- [1] Dr. Hab. Eng. Chwiej, T. *Project: Molecular dynamics simulations in 2D*. AGH University of Krakow. Retrieved from https://galaxy.agh.edu.pl/~chwiej/comp_phys/labs/8_molecular_dynamics__2d.pdf